Arjuna JEE 2027

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

TRIGONOMETRIC EQUATIONS

DIBY (DO IT BY YOURSELF)

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DIBY-01

Solve : $\sqrt{3}\sec 2\theta = 2$.

[Ans. : $\theta = n\pi \pm \frac{\pi}{12}, n \in I$]

DIBY-02

Find general solution of equation $\cot x = \sqrt{3}$

(A) $x = n\pi + \frac{\pi}{3}$

(B) $x = n\pi + \frac{\pi}{6}$

(C) $x = n\pi + \frac{2\pi}{3}$

(D) $x = \frac{9\pi}{6}\frac{5\pi}{6}$

[Ans. : (B)]

DIBY-03

The number of values of x in the interval $[0, 5\pi]$ satisfying the equation $3\sin^2 x - 7\sin x + 2 = 0$ is:

(A) 0 (B) 2 (C) 6 (D) 8

[Ans. : (C)]

DIBY-04

General solution of the equation $\left(2\sin\theta - \sqrt{3}\right)^2 + \left(\tan\theta + \sqrt{3}\right)^2 = 0$

[Ans. : $(6n+2)\frac{\pi}{3}, n \in I$]

DIBY-05

Find the principal solution of $\sin 2x = -\frac{1}{2}$

(A) $\frac{7\pi}{6}, \frac{11\pi}{6}$

(B) $\frac{7\pi}{12}, \frac{11\pi}{12}, \frac{15\pi}{12}, \frac{19\pi}{12}$

(C) $\frac{7\pi}{12}, \frac{11\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}$

(D) None of these

[Ans. : (C)]

DIBY-06

Find the number of solution $\tan 3x = 1$ in the interval $x \in \left[-\frac{2\pi}{3}, \frac{5\pi}{6}\right]$.

[Ans. : 05]

DIBY-07

Find the number of solution of the equation $16^{\sin^2 x} + 16^{\cos^2 x} = 10$, where $0 \le x \le 2\pi$.

[Ans. : 08]

DIBY-08

Number of values of 'x' in $(-2\pi, 2\pi)$ satisfying the equation $2^{\sin^2 x} + 4(2^{\cos^2 x}) = 6$ is:

(A) 8 (B) 6 (C) 4 (D) 2

[Ans. : (C)]

DIBY-09

The most general solution of $\tan\theta = -1$ and $\cos\theta = \frac{1}{\sqrt{2}}$ is

(A) $n\pi + \frac{7\pi}{4}, n \in I$

(B) $n\pi + (-1)^n \frac{7\pi}{4}, n \in I$

(C) $2n\pi + \frac{7\pi}{4}, n \in I$

(D) None of these

[Ans. : (C)]

DIBY-10

If $4\sin^2\theta + 2(\sqrt{3}+1)\cos\theta = 4+\sqrt{3}$, where $n \in I$ then the general value of θ is

(A) $2n\pi \pm \frac{\pi}{3}$

(B) $2n\pi + \frac{\pi}{4}$

(C) $n\pi \pm \frac{\pi}{3}$

(D) $n\pi - \frac{\pi}{3}$

[Ans. : (A)]

DIBY-11

Solve for x : $\sin x\left(\cos\frac{x}{4} - 2\sin x\right) + \left(1 + \sin\frac{x}{4} - 2\cos x\right)\cos x = 0$

[Ans. : $2\pi(4n+1)$]

DIBY-12

The equation $\sqrt{3}\sin x + \cos x = 4$ has

(A) Only one solution

(B) Two solutions

(C) Infinitely many solutions

(D) No solution

[Ans. : (D)]

DIBY-13

The number of solutions of $\sin\theta + 2\sin 2\theta + 3\sin 3\theta + 4\sin 4\theta = 10$ in $(0, \pi)$ is

[Ans. : .....]

DIBY-14

The general solution of $8\tan^2\frac{x}{2} = 1 + \sec x$ is

(A) $x = 2n\pi \pm \cos^{-1}\left(\frac{-1}{3}\right)$

(B) $x = 2n\pi \pm \frac{\pi}{6}$

(C) $x = 2n\pi \pm \cos^{-1}\left(\frac{1}{3}\right)$

(D) None of these

[Ans. : (C)]

DIBY-15

Number of solutions of $\sum_{r=1}^{5} \cos rx = 5$ in the interval $[0, 4\pi]$ is:

(A) 0 (B) 2 (C) 3 (D) 7

[Ans. : (C)]

DIBY-16

$\frac{\sin 3\theta}{2\cos 2\theta+1} = \frac{1}{2}$ if

(A) $\theta = n\pi + \frac{\pi}{6}, n \in I$

(B) $\theta = 2n\pi - \frac{\pi}{6}, n \in I$

(C) $\theta = n\pi + (-1)^n \frac{\pi}{6}, n \in I$

(D) $\theta = n\pi - \frac{\pi}{6}, n \in I$

[Ans. : (C)]

DIBY-17

If $0 \leq x < \frac{\pi}{2}$, then the number of values of x for which $\sin x - \sin 2x + \sin 3x = 0$, is

(A) 2 (B) 1 (C) 3 (D) 4

[Ans. : (A)]

DIBY-18

If the sum of solutions of the system of equations $2\sin^2\theta - \cos 2\theta = 0$ and $2\cos^2\theta + 3\sin\theta = 0$ in the interval $[0, 2\pi]$ is $k\pi$, then k is equal to .

[Ans. : 03]

DIBY-19

Solve : $\sin(\cos x) = \cos(\sin x)$

[Ans. : No Solution]

DIBY-20

Solve : $\operatorname{cosec} x - \operatorname{cosec} 2x = \operatorname{cosec} 4x$

[Ans. : $x = (2n+1)\frac{\pi}{7}$ & $n \neq 7m+3, n \in I$ & $m \in I$]

DIBY-21

The general value of θ satisfying the equation $2\sin^2\theta - 3\sin\theta - 2 = 0$, is

[JEE-1995, 2M]

(A) $n\pi + (-1)^n \frac{\pi}{6}$

(B) $n\pi + (-1)^n \frac{\pi}{2}$

(C) $n\pi + (-1)^n \frac{5\pi}{6}$

(D) $n\pi + (-1)^n \frac{7\pi}{6}$

[Ans. : (D)]

DIBY-22

The general solution of $\sin x - 3\sin 2x + \sin 3x = \cos x - 3\cos 2x + \cos 3x$ is

[JEE-1989, 2M]

(A) $n\pi + \frac{\pi}{8}$ (B) $\frac{n\pi}{2} + \frac{\pi}{8}$ (C) $(-1)^n \frac{n\pi}{2} + \frac{\pi}{8}$ (D) $2n\pi + \cos^{-1}\frac{3}{2}$

[Ans. : (C)]

DIBY-23

The possible values of $\theta \in (0, \pi)$ such that $\sin(\theta) + \sin(4\theta) + \sin(7\theta) = 0$ are

[AIEEE-2011]

(A) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(B) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{4\pi}{9}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(C) $\frac{\pi}{4}, \frac{5\pi}{12}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(D) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{35\pi}{36}$

[Ans. : (B)]

DIBY-24

The number of distinct solutions of the equation

$\frac{5}{4}\cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$ in the interval $[0, 2\pi]$ is

[JEE (Adv.)-2015]

[Ans. : 08]

DIBY-25

The number of solutions of the equation $\sin x = \cos^2 x$ in the interval $(0, 10)$ is

[JEE Mains-2022]

[Ans. : 04]

DIBY-26

Consider the following lists : **[JEE (Adv.)-2022]**

List-I		List-II	
(I)	$\left\{x \in \left[-\frac{2\pi}{3}, \frac{2\pi}{3}\right] : \cos x + \sin x = 1\right\}$	(P)	has two elements
(II)	$\left\{x \in \left[-\frac{5\pi}{18}, \frac{5\pi}{18}\right] : \sqrt{3}\tan 3x = 1\right\}$	(Q)	has three elements
(III)	$\left\{x \in \left[-\frac{6\pi}{5}, \frac{6\pi}{5}\right] : 2\cos(2x) = \sqrt{3}\right\}$	(R)	has four elements
(IV)	$\left\{x \in \left[-\frac{7\pi}{4}, \frac{7\pi}{4}\right] : \sin x - \cos x = 1\right\}$	(S)	has five elements
		(T)	has six elements

The correct option is:

(A) (I)→(P); (II)→(S); (III)→(P); (IV)→(S) (B) (I)→(P); (II)→(P); (III)→(T); (IV)→(R)

(C) (I)→(Q); (II)→(P); (III)→(T); (IV)→(S) (D) (I) →(Q); (II) →(S); (III) →(P); (IV) →(R)

[Ans. : (B)]

DIBY-27

If the solution of the equation $\log_{\cos x}\cot x + 4\log_{\sin x}\tan x = 1, x \in \left(0, \frac{\pi}{2}\right)$, is $\sin^{-1}\left(\frac{\alpha+\sqrt{\beta}}{2}\right)$, where α, β are integers, then $\alpha + \beta$ is equal to: [JEE Mains-2023]

(A) 3 (B) 5 (C) 6 (D) 4

[Ans. : (D)]

DIBY-28

Let S be the set of all $\alpha \in R$ such that the equation, $\cos 2x + \alpha \sin x = 2\alpha - 7$ has a solution. Then S is equal to: [JEE Mains-2019]

(A) $[2, 6]$ (B) $[3, 7]$ (C) R (D) $[1, 4]$

[Ans. : (A)]

DIBY-29

If $\alpha, -\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ is the solution of $4\cos\theta + 5\sin\theta = 1$, then the valuer of $\tan\alpha$ is

[JEE Mains-2024]

(A) $\frac{10-\sqrt{10}}{6}$ (B) $\frac{10-\sqrt{10}}{12}$ (C) $\frac{\sqrt{10}-10}{12}$ (D) $\frac{\sqrt{10}-10}{6}$

[Ans. : (C)]

DIBY-30

The sum of all values of $\theta \in \left(0, \frac{\pi}{2}\right)$ satisfying $\sin^2 2\theta + \cos^4 2\theta = \frac{3}{4}$ is:

[JEE Mains-2019]

(A) π (B) $\frac{5\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $\frac{3\pi}{8}$

[Ans. : (C)]

DIBY-31

Let $S = \{\theta \in [-2\pi, 2\pi] : 2\cos^2 \theta + 3\sin \theta = 0\}$. Then the sum of the elements of S is

[JEE Mains-2019]

(A) $\frac{13\pi}{6}$ (B) π (C) 2π (D) $\frac{5\pi}{2}$

[Ans. : (C)]

DIBY-32

Let the set of all $a \in R$ such that the equation $\cos 2x + a\sin\ x = 2a - 7$ has a solution be $[p, q]$ and $r = \tan 9^\circ - \tan 27^\circ - \frac{1}{\cot 63^\circ} + \tan 81^\circ$, then pqr is equal to

[JEE Mains-2024]

[Ans. : 48]

DIBY-33

Solve: $\sin\theta + \sqrt{3}\cos\theta \geq 1, -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

[Ans. : $-\frac{\pi}{6} \leq \theta \leq \frac{\pi}{2}$]

DIBY-34

Find the value of x in the interval $\left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$ for which $\sqrt{2}\sin 2x + 1 \leq 2\sin x + \sqrt{2}\cos x$

[Ans. : $x \in \left[\frac{-\pi}{4}, \frac{\pi}{6}\right] \cup \left[\frac{\pi}{4}, \frac{5\pi}{6}\right]$]

DIBY-35

Solve $\sin 2x > \sqrt{2}\sin^2 x + (2 - \sqrt{2})\cos^2 x$

[Ans. : $\left(n\pi + \frac{\pi}{8}, n\pi + \frac{\pi}{4}\right), n \in I$]

DIBY-36

The set of all x in $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$ satisfying $|4\sin x - 1| < \sqrt{5}$ is given by

(A) $\left(-\frac{\pi}{10}, \frac{3\pi}{10}\right)$ (B) $\left(\frac{\pi}{10}, \frac{-3\pi}{10}\right)$ (C) $\left(\frac{\pi}{10}, \frac{3\pi}{10}\right)$ (D) None of these

[Ans. : (A)]

DIBY-37

Solve for x & y :

$x\cos^3 y + 3x\cos y\sin^2 y = 14$

$x\sin^3 y + 3x\cos^2 y\sin y = 13$

[Ans. : $x = \pm 5\sqrt{5}$ & $y = n\pi + \tan^{-1}\frac{1}{2}$]

DIBY-38

Solve $\left(\sin^2 x + \frac{1}{\sin^2 x}\right)^2 + \left(\cos^2 x + \frac{1}{\cos^2 x}\right)^2 = 12 + \frac{1}{2}\sin y$

[Ans. : $x = k\pi \pm \frac{\pi}{4}; y = 2n\pi + \frac{\pi}{2}; k, n \in I$]

DIBY-39

Solve x and y : $4^{\sin x} + 3^{\frac{1}{\cos y}} = 11$

$5.16^{\sin x} - 2.3^{\frac{1}{\cos y}} = 2$

[Ans. : $x = n\pi + (-1)^n\frac{\pi}{6}$ and $y = 2m\pi \pm \frac{\pi}{3}$, where $m, n \in I$]

DIBY-40

Consider the trigonometric equation : $\cos 4x + 6 = 7\cos 2x$.

Which of the following represents the correct general solution for x, and what is the sum of all solution in the interval $[0, 314]$?.

(A) General Solution: $x = n\pi$; Sum of solutions: 4950π

(B) General Solution: $x = n\pi$; Sum of solutions: 5050π

(C) General Solution: $x = n\pi$; Sum of solutions: 5000π

(D) General Solution: $x = n\pi$, Sum of solutions: 4900π

[Ans. : (A)]

DIBY-41

The number of real solutions to the trigonometric equation $2\sqrt{3}\cos x + \sin x = 2$ in the interval $[0, 2\pi]$ is :

(A) 0 (B) 1 (C) 2 (D) 4

[Ans. : (C)]

DIBY-42

The general solution of the trigonometric equation $(\sec x - 1) = (\sqrt{2} - 1)\tan x$ is (where $n \in \mathbb{Z}$):

(A) $x = 2n\pi$ or $x = 2n\pi + \frac{\pi}{4}$

(B) $x = 2n\pi$ or $x = 2n\pi + \frac{\pi}{8}$

(B) $x = n\pi$ or $x = n\pi + \frac{\pi}{8}$

(D) $x = n\pi$ or $x = n\pi + \frac{\pi}{4}$

[Ans. : (A)]

DIBY-43

Find the general solution and number of solutions of the equation $\cos^2 x + \cos^2 2x + \cos^2 3x + \cos^2 4x = 2$ in the interval $[0, \pi]$.

(A) 5 (B) 7 (C) 8 (D) 10

[Ans. : (B)]

DIBY-44

Find the general solution and number of solutions of the equation $\sin^4 2x + \cos^4 2x = \sin 2x \cdot \cos 2x$ in the interval $[0, 2\pi]$.

(A) 2 (B) 4 (C) 6 (D) 8

[Ans. : (B)]

DIBY-45

Find the general solution and number of solutions of the equation $\text{cosec}x - \text{cosec}2x = \text{cosec}4x$ in the interval $[0, 2\pi]$.

(A) 4 (B) 7 (C) 8 (D) 6

[Ans. : (D)]

DIBY-46 (Challenger Question)

If $0 < \alpha, \beta < \pi$ and $\cos\alpha + \cos\beta - \cos(\alpha + \beta) = \frac{3}{2}$, find α and β.

[Ans. : $\alpha = \beta = \frac{\pi}{3}$]

DIBY-47 (Critical Thinking Question)

Find the solution set of the equation, $\log_{\frac{-x^2-6x}{10}} (\sin 3x + \sin x) = \log_{\frac{-x^2-6x}{10}} (\sin 2x)$

[Ans. : $-\frac{5\pi}{3}$]

DIBY-48

The number of solutions of the pair of equations

$2\sin^2\theta - \cos 2\theta = 0$

$2\cos^2\theta - 3\sin\theta = 0$ in the interval $[0, 2\pi]$ is [IIT-JEE-2007, Paper-1]

(A) Zero (B) One (C) Two (D) Four

[Ans. : (C)]

DIBY-49

Let $P = \{\theta : \sin\theta - \cos\theta = \sqrt{2}\cos\theta\}$ and $Q = \{\theta : \sin\theta + \cos\theta = \sqrt{2}\sin\theta\}$ be two sets. Then [IIT-JEE-2011]

(A) $P \subset Q$ and $Q - P \neq \varnothing$ (B) $Q \not\subset P$

(C) $P \not\subset Q$ (D) $P = Q$

[Ans. : (D)]

DIBY-50

Let $S = \{\theta \in [-2\pi, 2\pi] : 2\cos^2\theta + 3\sin\theta = 0\}$. Then the sum of the elements of S is:

(A) π (B) $\frac{5\pi}{3}$ (C) $\frac{13\pi}{6}$ (D) 2π

[Ans. : (D)]

DIBY-51

Let α and β respectively be the maximum and the minimum values of the function $f(\theta) = 4\left(\sin^4\left(\frac{7\pi}{2} - \theta\right) + \sin^4(11\pi + \theta)\right) - 2\left(\sin^6\left(\frac{3\pi}{2} - \theta\right) + \sin^6(9\pi - \theta)\right)$, $\theta \in \mathbb{R}$ Then $\alpha + 2\beta$ is equal to: [JEE (Main)-2026]

(A) 4 (B) 5 (C) 3 (D) 6

[Ans. : (B)]

DIBY-52

The least value of $\left(\cos^2\theta - 6\sin\theta\cos\theta + 3\sin^2\theta + 2\right)$ is [JEE (Main)-2026]

(A) –1 (B) $4 + \sqrt{10}$ (C) $4 - \sqrt{10}$ (D) 1

[Ans. : (C)]

DIBY-53

The value of $\frac{\sqrt{3}\text{cosec}20^\circ - \sec 20^\circ}{\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ}$ is equal to [JEE (Main)-2026]

(A) 32 (B) 16 (C) 64 (D) 12

[Ans. : (C)]

DIBY-54

If $\frac{\tan(A-B)}{\tan A}+\frac{\sin^2 C}{\sin^2 A}=1, A, B, C \in \left(0, \frac{\pi}{2}\right)$, then [JEE (Main)-2026]

(A) $\tan A, \tan C, \tan B$ are in G.P.

(B) $\tan A, \tan B, \tan C$ are in G.P.

(C) $\tan A, \tan C, \tan B$ are in A.P.

(D) $\tan A, \tan B, \tan C$ are in A.P.

[Ans. : (A)]

DIBY-55

The number of solutions of the equation $\sin(e^x)=5^x+5^{-x}$ is [IIT-JEE 1991]

(A) 0

(B) 1

(C) 2

(D) Infinitely many

[Ans. : (A)]

DIBY-56

The equation $(\cos p-1)x^2+(\cos p)x+\sin p=0$ in the variable x, has real roots. Then, p can take any value in the interval [IIT-JEE 1990]

(A) $(0, 2\pi)$

(B) $(-\pi, 0)$

(C) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(D) $(0, \pi)$

[Ans. : (D)]

DIBY-57

Let $S = \{x \in [-\pi, \pi] : \sin x(\sin x + \cos x) = a, a \in Z\}$. Then $n(S)$ is equal to:

[JEE (Main)-2026]

(A) 3 (B) 6 (C) 7 (D) 9

[Ans. : (D)]

DIBY-58

The smallest positive root of the equation $\tan x - x = 0$ lies in [IIT-JEE 1987]

(A) $\left(0, \frac{\pi}{2}\right)$ (B) $\left(\frac{\pi}{2}, \pi\right)$ (C) $\left(\pi, \frac{3\pi}{2}\right)$ (D) $\left(\frac{3\pi}{2}, 2\pi\right)$

[Ans. : (C)]

DIBY-59

For all θ in $[0, \pi/2]$, show that $\cos(\sin\theta) > \sin(\cos\theta)$. [IIT-JEE 1981]

DIBY-60

Find the smallest positive number p for which the equation $\cos(p\sin x) = \sin(p\cos x)$ has a solution $x \in [0, 2\pi]$. [IIT-JEE 1995]

[Ans. : $p = \frac{\pi}{2\sqrt{2}}$]

DIBY-61

Show that the equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has no real solution. [IIT-JEE 1982]

It is never about end result

It's always about Journey

Thank You

#futureIITians